

Erick Vieyra

Software Engineer

sevieyra@gmail.com
[linkedin.com/in/sevieyra](https://www.linkedin.com/in/sevieyra)
San Diego, California

Technical skills

Systems and Software:

Modern C++ (14/17/20), Python, Linux
Kernel, RTOS (VxWorks), Multi-threading,
Performance Tuning (GDB, Valgrind, Perf),

Cloud and Infrastructure

Docker, CI/CD Architecture, Scalable
Monitoring and Observability,
Infrastructure-as-Code.

HPC & Distributed Systems

NVIDIA CUDA, Latency-Critical
Design, ROS2, RTI Connex DDS,
Middleware Architecture.

Work Experience

Software Engineer, Firmware Tools and Infrastructure - Zoox

2022 - Today. San Diego, CA

Project Lead for the end-to-end architecture of vehicle middleware simulation and Hardware-in-the-Loop (HIL) infrastructure. Improved testing and simulation workflows across the firmware organization, resulting in a measurable increase in release velocity, testability, and CI infrastructure reliability.

- Worked as a tech lead and responsible for our vehicle's middleware testing and simulation infrastructure.
- Worked with stakeholders and the rest of the team to add capabilities to Zoox's internal testing framework.
- Improved observability of our validation tooling Clickhouse and Grafana.
- Created and maintained tooling for integrating Pytest with our Hardware in the loop (HIL) testing equipment.
- Collaborate cross-functionally within the organization to set up Hardware in the Loop assets used for testing several ECUs.
- Mentored junior engineers on our internal tooling and framework best practices.

Software Engineer, Embedded Systems - TuSimple

2021 - 2022. San Diego, CA

GPU-accelerated Embedded Systems development of mission-critical, real-time autonomous driving system

- Worked with CUDA kernels for real-time Lidar point cloud processing on the NVIDIA Orin platform.
- Integrated Hesai Lidar drivers with NVIDIA DriveWorks framework, implementing custom decoders to translate raw UDP packets into GPU-accessible point cloud formats.
- Developed non-GPU, Autosar compliant C++ code for our autonomous truck.
- Revamped the testing and validation effort for the TuSimple Sensor Unit (TSU) by leveraging custom-made HIL setups.

Software Engineer, Robotics - Advanced Robotics Group

2019 - 2021. Houston, TX

"Robotics-Full-Stack" role encompassing embedded systems programming, ROS development, computer vision, big data visualization, and web development for a mission-critical, mobile robotics platform. Fourth engineering hire in the company.

- Modernized a ROS codebase using modern C++ features and tooling.
- Worked with Computer Vision algorithms and wrote drivers for interfacing with GigE and USB Cameras.
- Worked on detection tasks using cameras and other sensors using OpenCV.
- Created a big data visualization pipeline using Python, Jupyter Notebooks, Numpy, Matplotlib, and Dataslayer.
- Prototyped robot algorithms using Python and Rospy.
- Interfaced sensors using I2C and other IC communication protocols
- Created a CI/CD pipeline for the codebase using Github actions.
- Worked on a backend application of a customer portal using Django.
- Created a client-facing frontend application using Typescript, React, Redux and DeckGL.
- Implemented Code Reviews, Bug Tracking, daily standups and other standard software development practices.

Software Engineer, Embedded Systems, ABB Robotics

2017 - 2019. Västerås, Sweden

Embedded software development within ABB's industrial robot controllers. Part-time (25%) certified SCRUM Master.

- C++ development on the Operating System Abstraction Layer (OSAL) of ABB's robot controller.
- Worked with VxWorks, as well as Windows synchronization features.
- Part-time SCRUM master responsibilities.
- Worked on refactoring and maintenance of ABB's codebase.
- Worked on the new generation of ABB robot controllers.
- Created from scratch a 'Recovery Mode Web Application' using a C++ backend and VueJS as a frontend.
- Collaborated with ABB software developers from all around the world.



Academic Formation

Mälardalen University

2014 - 2016. Västerås, Sweden

Msc in Intelligent Embedded Systems, Academic scholarship holder.

Relevant courses/projects:

- Project in Autonomous Robotics. 30 ECTS: Collaboration project between Mälardalen University and Volvo with the end goal of building an autonomous garbage collecting robot. My contributions were mainly in deploying the SLAM algorithm using the ROS gmapping package.
- Project in Intelligent Embedded Systems. 30 ECTS: Collaboration project between Mälardalen University and ABB Robotics for deploying a fast object detection algorithm for pick-and-place applications. OpenCV implemented in a WR Linux embedded system.
- Research Work in Industrial IoT at ABB Robotics (Thesis Project): A research and development project with the goal to integrate several industrial robot controllers within a local cloud in order to enhance their capabilities. Worked with ROS (*roscpp* and *rospy*), and integrated it with ABB's RobotStudio. RTI Connex DDS. Research work with ROS2 (at that time in alpha stage) and integration with ABB robots.

Tecnológico de Monterrey

2009 - 2013. CDMX, México

Bsc in Digital Electronics and Robotics, Honorable Mention and Academic Scholarship Holder.

Relevant courses/projects:

- Computer Vision: Math-oriented course in OpenCV about classic computer vision algorithms.
- Control Engineering: Classic continuous and discrete control engineering theory and practical exercises in Matlab.
- Industrial Robotics: Kinematics, motion control and path planning using Peter Corke's robotics toolbox for Matlab.



Certifications and Merits

Applied Data Science with Python Specialization

2020

Coursera / University of Michigan Data Science certification.

SAFe Scrum Master

2018

Certified Scrum Master by Scaled Agile Framework.

ABB Robotics Scholarship

2015

First prize for an academic project related to robotics user interaction.

Academic Scholarship - Mälardalen University

2014

Full academic scholarship for studying a master degree at Mälardalen University.

CENEVAL Award

2014

National award granted to the top achieving students during the national, standardized test for graduate students in Mexico.

Academic scholarship for studying a bachelor degree at Tecnológico de Monterrey, Mexico State Campus.